Light

GCSE Computer Science — Boolean Logic (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What output does an AND gate produce when at least one input is 0?

- A. 0
- B. 1
- C. The value of the other input
- D. Undefined

Q2 (1 mark)

What output does an OR gate produce when both inputs are 0?

- A. 0

- **B.** 1
- **C.** Either 0 or 1 randomly
- **D.** The gate does not function

Q3 (1 mark)

How many inputs does a NOT gate have?

- **A.** 0 inputs
- **B.** 1 input
- **C.** 2 inputs
- **D.** 3 inputs

Q4 (1 mark)

Which logic gate symbol has a flat back (input side) and a curved, rounded front (output side)?

- **A.** AND gate
- **B.** OR gate
- **C.** NOT gate
- **D.** No standard gate has this shape

Q5 (1 mark)

How many rows does a truth table have for a logic circuit with 2 inputs?

- ○ **A.** 2 rows
- ○ **B.** 3 rows
- ○ **C.** 4 rows
- ○ **D.** 8 rows

Q6 (1 mark)

In some truth tables, T and F are used instead of 1 and 0. What do T and F represent?

- ○ **A.** T means 'toggle', F means 'fixed'
- ○ **B.** T and F are error codes
- ○ **C.** T means 'top', F means 'floor'
- ○ **D.** T means 'True' (equivalent to 1), F means 'False' (equivalent to 0)

Q7 (1 mark)

A lift only moves when the doors are closed (D=1) AND the floor button is pressed (B=1). Which gate models this safety system?

- ○ **A.** OR gate
- ○ **B.** NOT gate
- ○ **C.** AND gate
- ○ **D.** No gate is needed

Q8 (1 mark)

A fire sprinkler activates if the smoke detector triggers (S=1) OR the heat sensor triggers (H=1). Which gate represents this?

- ○ **A.** AND gate
- ○ **B.** OR gate
- ○ **C.** NOT gate
- ○ **D.** A combination of all three gates

Q9 (1 mark)

A burglar alarm is armed (output=1) when it is NOT disabled (disable switch=0). Which gate produces this behaviour?

- ○ **A.** AND gate
- ○ **B.** OR gate
- ○ **C.** NOT gate
- ○ **D.** No gate produces this behaviour

Q10 (1 mark)

What is the result produced by a logic gate called?

- ○ **A.** The input
- ○ **B.** The operand
- ○ **C.** The function

- ○ **D.** The output

Q11 (1 mark)

How many outputs does a basic logic gate (AND, OR, or NOT) produce?

- ○ **A.** 0 outputs
- ○ **B.** 1 output
- ○ **C.** 2 outputs
- ○ **D.** It varies depending on the gate type

Q12 (1 mark)

In a NOT gate truth table, what is the output when the input is 0?

- ○ **A.** 0
- ○ **B.** Undefined
- ○ **C.** Same as input
- ○ **D.** 1

Q13 (2 marks)

What is the output of NOT (1 OR 1)?

- ○ **A.** 0

- B. 1
- C. 2
- D. Cannot be determined

Q14 (2 marks)

What is the output of NOT (0 AND 0)?

- A. 0
- B. 1
- C. Depends on the circuit design
- D. -1

Q15 (2 marks)

Evaluate: (0 OR 1) AND (1 OR 0)

- A. 0
- B. 10
- C. 01
- D. 1

Q16 (2 marks)

Evaluate: (NOT 0) OR (NOT 1)

- **A.** 0
- **B.** 1
- **C.** Both 0 and 1
- **D.** Undefined

Q17 (2 marks)

In a truth table for A AND B, which row shows: A=1, B=0, Output=? What is the missing output value?

- **A.** 1
- **B.** B
- **C.** A
- **D.** 0

Q18 (2 marks)

In a truth table for A OR B, which row shows: A=1, B=0, Output=? What is the missing output value?

- **A.** 0
- **B.** Same as previous row
- **C.** Cannot be determined without more information
- **D.** 1

Q19 (2 marks)

A logic circuit is described as (NOT A) OR B. If A=1 and B=0, what is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** Depends on gate order
- ○ **D.** Cannot process these values

Q20 (2 marks)

Which Boolean expression outputs 1 when A=0 OR when B=1 (or both)?

- ○ **A.** A AND B
- ○ **B.** A OR B
- ○ **C.** (NOT A) OR B
- ○ **D.** A AND (NOT B)

Q21 (2 marks)

Evaluate: (A AND B) OR C, where A=0, B=1, C=1

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2

- D. Undefined

Q22 (2 marks)

NOT (A OR B) produces the same output as which expression when tested with all input combinations?

- A. (NOT A) AND (NOT B)
- B. (NOT A) OR (NOT B)
- C. A AND B
- D. A OR B

Q23 (2 marks)

A greenhouse ventilation fan activates when temperature is high (T=1) OR humidity is high (H=1), but only if the manual override is not active (M=0). Which expression models this?

- A. (T AND H) AND M
- B. (T OR H) AND (NOT M)
- C. (T OR H) OR M
- D. T AND H AND (NOT M)

Q24 (3 marks)

A logic circuit has two inputs X and Y. The complete truth table is:

X	Y	Output

0	0	0
0	1	0
1	0	0
1	1	1

Which Boolean expression produces this truth table?

- ○ **A.** X OR Y
- ○ **B.** X AND Y
- ○ **C.** NOT X OR Y
- ○ **D.** NOT (X AND Y)

Q25 (3 marks)

A logic circuit is: (NOT A) AND (B OR C).
For which input combination does the output equal 1?

- ○ **A.** A=1, B=0, C=0
- ○ **B.** A=1, B=1, C=0
- ○ **C.** A=0, B=0, C=0
- ○ **D.** A=0, B=1, C=0

Q26 (3 marks)

A gaming console has parental controls. A game can be played when:

- The user is an adult (A=1), OR
- The game is age-appropriate (G=1) AND parental PIN is entered (P=1)

Which expression correctly models this system?

- **A.** A AND G AND P
- **B.** A OR G OR P
- **C.** A OR (G AND P)
- **D.** (A OR G) AND P

Q27 (3 marks)

Complete the truth table for: (NOT A) OR B

A	B	Output
0	0	?
0	1	?
1	0	?
1	1	?

Which row pattern shows the correct outputs (in order from top to bottom)?

- **A.** 0, 0, 0, 1
- **B.** 1, 1, 0, 1
- **C.** 0, 1, 0, 1
- **D.** 1, 1, 1, 0

Q28 (3 marks)

A student designs a circuit for a quiz buzzer system. The buzzer should sound (output=1) when Player 1 buzzes (P1=1) OR Player 2 buzzes (P2=1).

The student incorrectly uses: P1 AND P2
What is wrong, and when does the student's circuit fail?

- ○ **A.** The circuit works correctly; there is no error
- ○ **B.** The circuit fails when only one player buzzes - it stays silent when P1=1, P2=0 or P1=0, P2=1
- ○ **C.** The circuit fails when both players buzz - it stays silent when P1=1, P2=1
- ○ **D.** The circuit produces random outputs

Q29 (3 marks)

A traffic light controller uses this logic for the green light:
Green = (Timer expired (T=1) AND No emergency vehicle (E=0)) OR Manual override (M=1)
Given T=1, E=1, M=0, is the light green?

- ○ **A.** Yes, because the timer has expired
- ○ **B.** Yes, because manual override is active
- ○ **C.** No, because an emergency vehicle is present
- ○ **D.** No, because manual override is not active

Q30 (3 marks)

A logic circuit uses the expression: (A OR B) AND (A OR NOT B). Analyse this expression. What simpler equivalent expression produces the same outputs?

- ○ **A.** A AND B
- ○ **B.** A OR B

- ○ **C.** A
- ○ **D.** B

Total: **55** marks • Time: **30 min**

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